



VAISHNAVEE BV

192571071

CSA1607 DATA WAREHOUSING AND DATA MINING

Behind Every Click: How Netflix Remembers You — A Journey Through Dimensional Data Modeling

Understanding Star, Snowflake & Galaxy Schemas through a real recommendation system.

For data analytics students and early-career data engineers — a visual, practical tour of the models that power fast recommendations and analytical dashboards.

Behind Every Click

How Netflix Remembers You

Imagine opening Netflix. Within seconds it knows what you'll watch, which genre you love, when you usually watch, and which actor keeps you hooked.



In one giant table?

Feels simple, but quickly becomes slow and messy at scale.



Many connected tables?

Normalized and organized — easier to maintain, more joins.



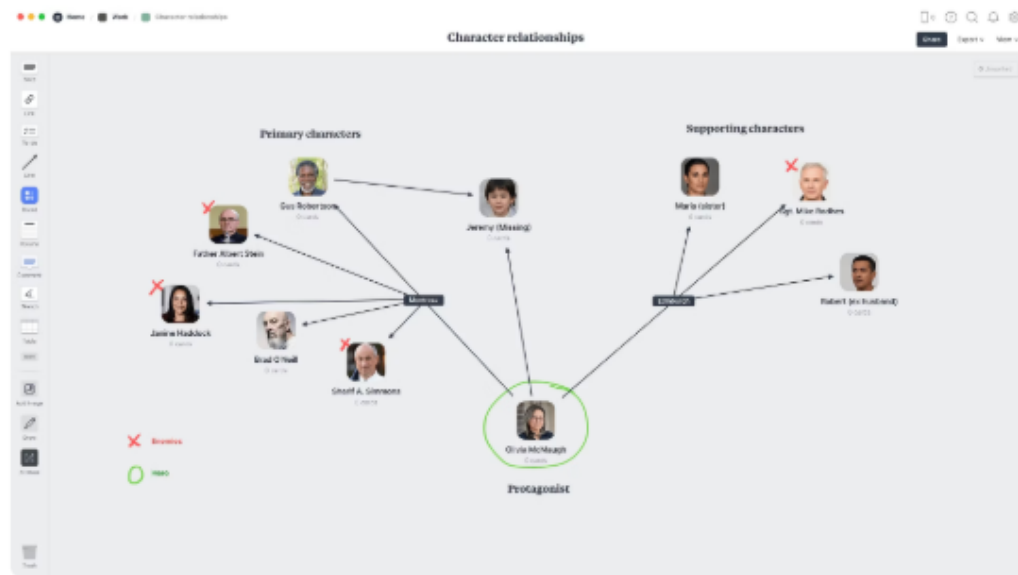
AI remembers everything?

AI helps predictions, but it relies on well-modeled data underneath.

Today we'll discover the database architecture behind systems like this and why dimensional models are the secret sauce for analytics and recommendations.

Turning Raw Data into Business Intelligence

- Instead of storing raw events like "Customer bought Movie A on Friday", we reshape data into facts (events) and dimensions (context) for fast analysis.



Key building blocks

- Facts — measurable events: play, purchase, rating
- Dimensions — descriptive context: Customer, Movie, Date, Location
- Goal — fast, intuitive queries with minimal complexity

Think: facts answer "what happened?"
dimensions answer "who/when/where/how?"

Star Schema — Fast Like Your Favorite Food Delivery App

The Star Schema centers a large fact table with multiple denormalized dimension tables radiating out. It's optimized for read performance and simple analytics.

Fact Table (central)

Contains keys and measures:
SalesID, ProductID, CustomerID,
Amount, Timestamp

Dimension Tables

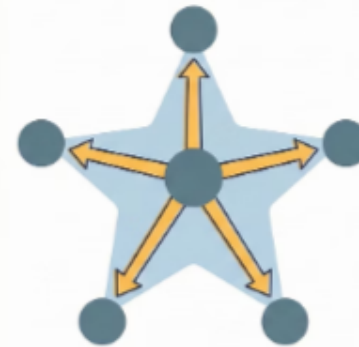
Descriptive attributes: Customer
(name, age, gender), Product/Movie
(title, genre), Date (day, hour)

Advantages

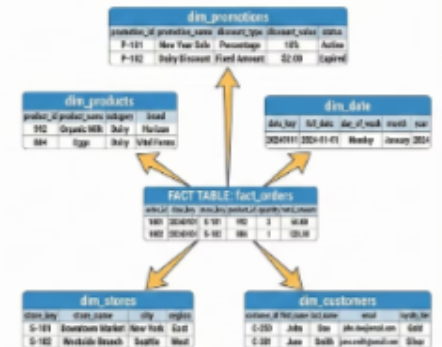
Fast queries, simple joins, ideal for dashboards and top-N queries (e.g., today's Top 10 Action Movies)

If Netflix wants today's Top 10 Action Movies, the Star Schema is designed to make that query fast — answer: YES.

STAR SCHEMA HUB AND SPOKE MODEL

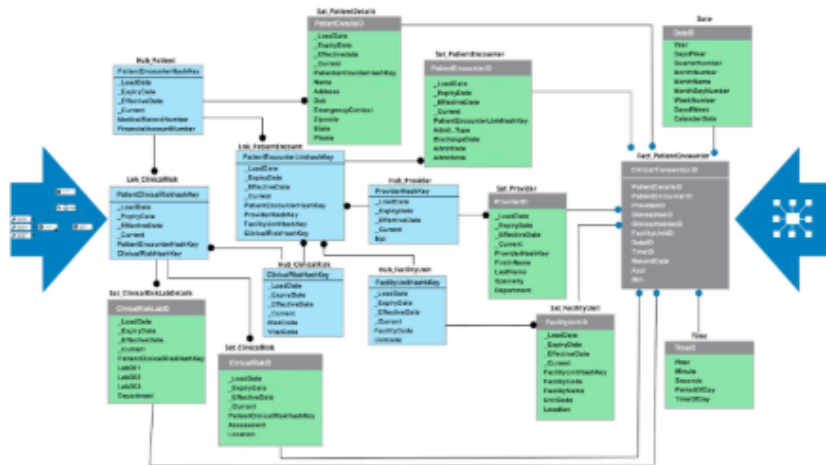


ACTUAL STAR SCHEMA WITH DATA



Snowflake Schema — Organized Like a Family Tree

Snowflake splits dimensions into related sub-tables (normalize attributes) to remove redundancy and enforce structure.



- Why normalize?

Less redundancy, clearer hierarchies (City → State → Country).

- Trade-offs

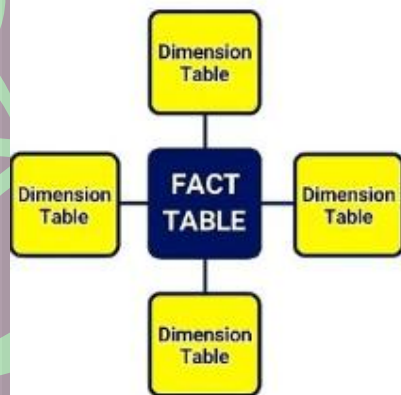
More joins and slightly slower query performance compared to Star.

- **When to use**

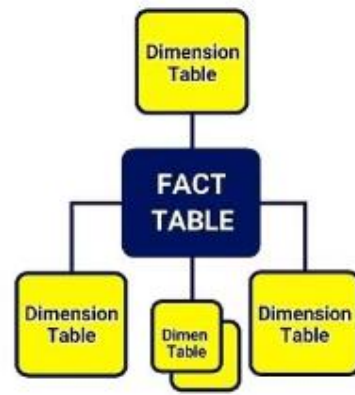
When storage constraints and strict data governance matter.

- ❏ Question: Which would you choose if storage is limited? (Snowflake favors storage efficiency.)

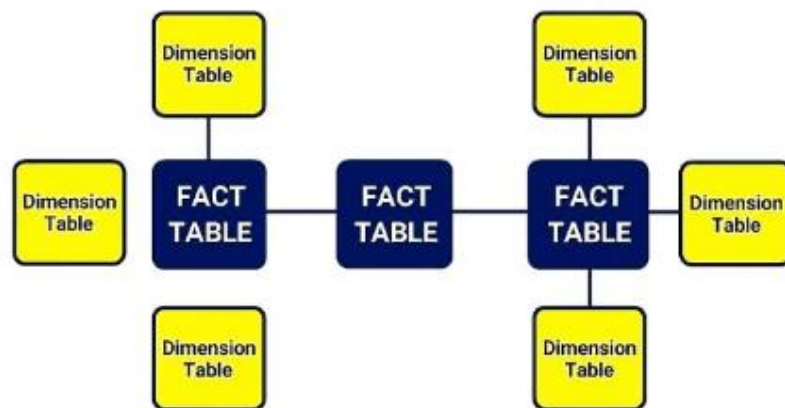
Types of Schema in Data Modeling



Star Schema ★



Snowflake Schema



Galaxy Schema — When One Business Isn't Enough

The Galaxy (or fact constellation) uses multiple fact tables that share dimension tables. It supports multiple processes and complex enterprise analytics.



Multiple Facts

Sales Fact, Returns Fact, Inventory Fact — separate facts for distinct events.



Shared Dimensions

Customer, Product, Date reused across facts for consistent analysis.



Enterprise Use

Ideal for Amazon, Walmart, hospitals, banks — complex cross-process queries.

Interactive: Want to analyze Sales + Returns together? Galaxy enables cross-fact joins using shared dimensions — powerful for enterprise analytics.

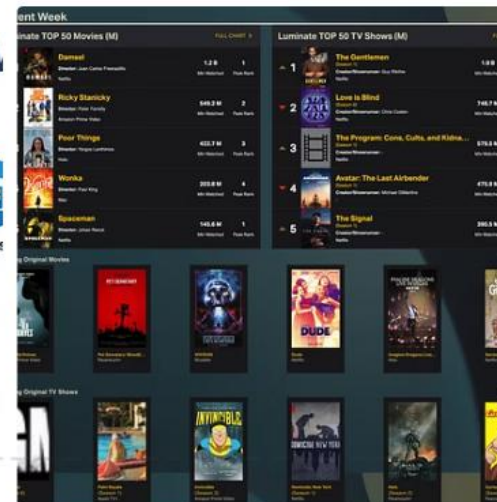
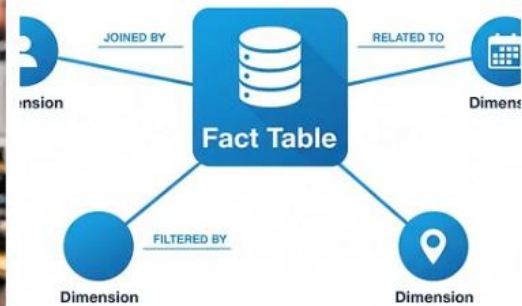
Which Schema Wins?

Feature	Star	Snowflake	Galaxy
Simplicity	★★★★★	★★★	★★
Speed	★★★★★	★★★	★★★★
Storage Efficiency	★★	★★★★★	★★★★
Scalability	★★★	★★★★	★★★★★
Best Use	BI Dashboards	Large Warehouses	Enterprise Analytics

Final Thought: Every recommendation on Netflix, every Amazon purchase, and every bank report begins with one smart data model. Choose based on your performance, storage, and scalability trade-offs.



SCHEMA CONCEPT M



☐ If you were building your startup today, which schema would you choose? ★ Star ❄ Snowflake 🌌 Galaxy